

30/07/26

Signal Processing

- Recap: Averaging system, Differencing system
- Notion of frequency: DTFT, DFT
- Analyse signals and systems in the frequency domain
- Topical outline:

- $$x[n] = \sum_{k=-\infty}^{\infty} a[k] \delta[n-k] \quad - (1)$$

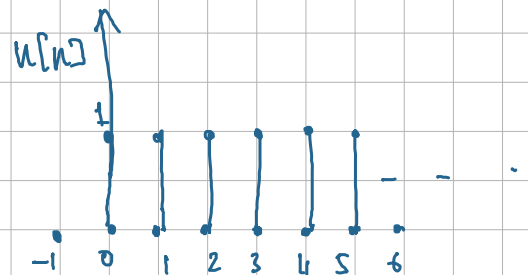
- $$x_{ball}[n] = 8 \cdot \delta[n] + 4 \delta[n-1] + 0 \delta[n-2] + \dots + 0 \delta[n-5]$$

- $$x_{over}[n] = 2 \cdot \delta[n] + 1 \delta[n-1] + 4 \delta[n-2] + \dots$$

- $$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

$$u[n] = \sum_{k=-\infty}^n \delta[k]$$

$$u[n] = \sum_{k=0}^{\infty} \delta[n-k]$$



- $$x[n] = e^{j\omega n} \quad \uparrow \text{common signal examples}$$

- Abstract system: $T\{x[n]\} = y[n]$

Example 1: $y[n] = x[n] - x[n-1]$

$$y'[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k]$$

verify if $h[n] \stackrel{\text{diff}}{=} \delta[n] - \delta[n-1]$

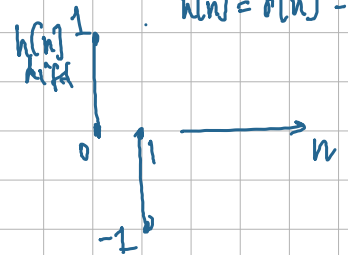
gives us the desired $y[n]$.

$$\begin{aligned} y[n] &= \sum_{k=-\infty}^{\infty} x[k] (\delta[n-k] - \delta[n-k-1]) \\ &= \sum_{k=-\infty}^{\infty} x[k] \cdot \delta[n-k] - \sum_{k=-\infty}^{\infty} x[k] \cdot \delta[n-k-1] \end{aligned}$$

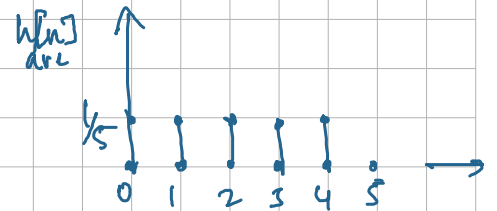
$$\left. \begin{aligned} y(t) &= \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau \end{aligned} \right\}$$

$$= \sum_{k=-\infty}^{\infty} x[k] \delta[n-k] - \sum_{k=-\infty}^{\infty} x[k] \delta[(n-1)-k]$$

$$h[n] = \delta[n] - \delta[n-1]$$



Example 2: $h_{ave}[n] = \frac{1}{5} \sum_{k=0}^4 \delta[n-k]$



Suppose $h[n]$ is a linear and time invariant system.

$$\uparrow \{x[n]\} = T \left\{ \sum_{k=-\infty}^{\infty} x[k] \delta[n-k] \right\}$$

$$= \sum_{k=-\infty}^{\infty} x[k] \cdot T\{\delta[n-k]\} \quad \text{due to linearity}$$

$$= \sum_{k=-\infty}^{\infty} x[k] h[n-k] \quad \text{due to time invariance}$$